

Life in temperate forest

The factor that most affects life in temperate forest is the variable food supply. At all levels—from the tree tops to the forest floor—the life cycles of forest animals living in temperate forest move in step with the seasons so that they produce their young when food is easiest to find. Life is relatively easy in spring and summer; but in winter, when the food supply is at its lowest, special adaptations are needed for survival.

Feeding

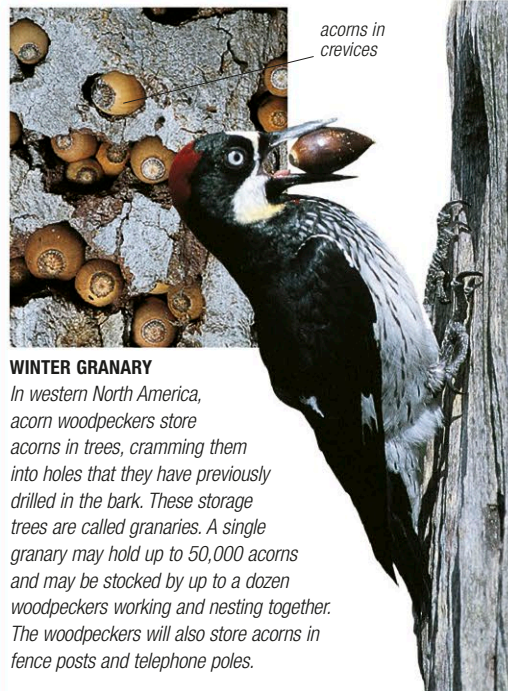
As a habitat, deciduous temperate forest—the kind of forest found across much of the northern hemisphere—has one very useful feature. The trees that grow in it, such as oak and beech, produce leaves that are designed to last for just one growing season. As a result, these leaves are usually thin and easy to eat, which is why vast numbers of insects feast on them from the moment they begin to appear in spring.

This sudden explosion of insect life attracts an army of highly specialized avian predators. In Europe, northern Asia, and North America, dozens of warbler species migrate north as the buds open. These birds have extremely acute eyesight, enabling them to scour leaves for the tiniest grubs and caterpillars, which they then pick up with their tweezerlike beaks. Other birds, including treecreepers, woodpeckers, and nuthatches, concentrate on the bark, seeking out and pecking at the tiny animals hidden among the crevices.

By midsummer, leaves stop growing and animal feeding behavior changes. Most temperate trees are pollinated by wind, which

FOREST-FLOOR FORAGERS

Watching over their piglets, adult wild boars root in the leaf litter for food. Wild boars feed on acorns and other nuts, and they also use their spadelike snouts to dig up roots, fungi, and small animals hidden in soil or among fallen leaves.



WINTER GRANARY

In western North America, acorn woodpeckers store acorns in trees, cramming them into holes that they have previously drilled in the bark. These storage trees are called granaries. A single granary may hold up to 50,000 acorns and may be stocked by up to a dozen woodpeckers working and nesting together. The woodpeckers will also store acorns in fence posts and telephone poles.

means that they do not produce enticing nectar-rich flowers. However, these trees do produce large crops of nuts and other seeds, which are extremely important foods for animals because, unlike leaves, they can be stored away and used when other food is scarce.

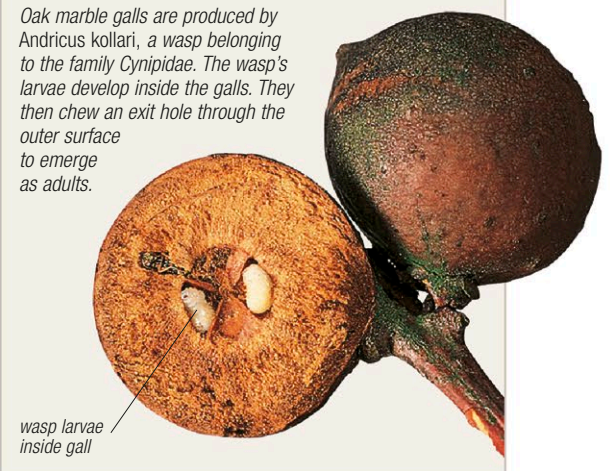
Food storage, or “caching,” is practiced by many forest birds and mammals. Jays bury acorns in the ground, while acorn

TREE GALLS

Common in broad-leaved forest, galls are growths whose development is triggered by insects and other organisms. Galls often look like berries or buttons on leaves and twigs. They act as both home and larder, or nursery, allowing their occupants to feed unseen. In broad-leaved forest, most galls are produced by tiny wasps and flies. Each species of insect affects a particular tree, and each produces galls of a characteristic shape, making them easy to identify.

OAK GALLS

Oak marble galls are produced by *Andricus kollari*, a wasp belonging to the family Cynipidae. The wasp's larvae develop inside the galls. They then chew an exit hole through the outer surface to emerge as adults.



woodpeckers store them in trees. Squirrels bury seeds of all kinds, and red foxes bury anything that is even faintly edible, from half-eaten remains to food wrappers and discarded shoes. Some animals locate their stores by scent, but most are very good at pinpointing them by memory alone, finding and digging up their food even when it is covered by snow.

Seed-caching has an important impact on forest ecology. Although animals that bury seeds have good memories, some of what they hide is always forgotten about. This means that provided the seeds are not discovered by rodents or other animals, they remain effectively planted and ready to germinate, helping the forest's trees to reproduce.

